

Data Flow Diagram

Maximum Marks: 2 Marks

Date	Team ID	Project Name
15 March 2026	Batch 2023-27 ACOE	Rising Waters – ML-Based Flood Prediction System

Team Members

S.No	Team Member Name	Role
1	Venkatesh Balireddy	Team Lead
2	Archana Dhanani	Member
3	Shivatmika Gandikota	Member
4	Gode Siva Ramakrishna Durgaprasad	Member
5	Bavisetty Gopi Krishna	Member

Level 0 — Context Diagram

The context diagram shows the Rising Waters system boundary, its external entities, and the primary data flows entering and leaving the system.

Element	Description
External Entity	Disaster Management Officer / Meteorologist
System	Rising Waters Flood Prediction System (Flask + XGBoost)
Input Flow	annual_rainfall, seasonal_rainfall, cloud_visibility, humidity, temperature, river_level
Output Flow	Binary prediction label (Flood / No Flood) + probability score + emergency protocol

Level 1 — Internal Data Flow

Shows how data moves between internal processes within the Rising Waters system.

Process ID	Process Name	Input	Output	Data Store
P1	Form Input Capture	User web form values	Raw feature dictionary	—
P2	Input Validation	Raw feature dict	Validated dict / Error msg	—
P3	Feature Scaling	Validated features	Scaled NumPy array	scaler.save

Process ID	Process Name	Input	Output	Data Store
P4	ML Prediction	Scaled NumPy array	Label (0/1) + Probability	floods.save
P5	Result Routing	Prediction label	HTTP redirect to result page	—
P6	Prediction Logging	All inputs + result + time	New row appended	predictions_log.csv